PHISHING WEBSITE DETECTION SYSTEM

Project Report

Course Code: CSA2001(B11+B12+B13)
Course Title: Fundamentals of AI and ML

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1. INTRODUCTION

Phishing is a type of cyber attack where attackers create fake websites that resemble legitimate ones in order to steal sensitive information such as usernames, passwords, and banking details. With the increasing reliance on the internet, phishing attacks have become more frequent and sophisticated.

This project focuses on developing a machine learning-based system that can automatically detect phishing websites based on specific features extracted from URLs and web pages.

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2. PROBLEM STATEMENT

Many internet users are unable to distinguish between legitimate and phishing websites. Traditional security measures are often not sufficient to detect newly emerging phishing techniques.

The problem addressed in this project is to build an intelligent system that can classify websites as legitimate, suspicious, or phishing using machine learning techniques.

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3. OBJECTIVES

- To develop a machine learning model for phishing detection
- To analyze and preprocess a real-world dataset
- To evaluate the performance of the model
- To create an interactive system for user input and prediction

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4. DATASET DESCRIPTION

The dataset used in this project is obtained from Kaggle. It contains 30 features that describe various characteristics of websites.

Examples of features include:

- SSL Final State
- URL of Anchor
- Web Traffic
- Having Subdomain
- Links in Tags
- Prefix/Suffix in Domain

The target variable is:
Result:
1 = Legitimate
0 = Suspicious
-1 = Phishing

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5. METHODOLOGY

5.1 Data Loading
The dataset is loaded using Pandas.

5.2 Data Preprocessing
The dataset is split into features (X) and target variable (y). Unnecessary columns such as 'key' are removed.

5.3 Train-Test Split
The dataset is divided into training and testing sets using an 80:20 ratio.

5.4 Model Selection
A Decision Tree Classifier is selected due to its simplicity and interpretability.

5.5 Model Training
The model is trained using the training dataset.

5.6 Model Evaluation
The model is evaluated using accuracy score on the test dataset.

5.7 Model Saving
The trained model is saved using Joblib for future use.

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6. IMPLEMENTATION

The system is implemented in Python using libraries such as Pandas, NumPy, and Scikit-learn.

The program performs the following steps:

- Loads dataset
- Trains the model
- Evaluates accuracy
- Accepts user input
- Predicts output

An interactive interface allows users to enter values for key features and receive predictions.

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7. RESULTS

The model achieved an accuracy of approximately 95.76% on the test dataset.

The system successfully classifies websites into:

- Safe
- Suspicious
- Phishing

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8. DISCUSSION

The Decision Tree model performs well on the dataset and provides good accuracy. However, the system depends on manual feature input, which may not be practical in real-world applications.

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9. LIMITATIONS

- Manual input required
- Limited to dataset features
- No real-time URL parsing
- Decision Tree may overfit

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10. FUTURE WORK

- Integration with web browser or API
- Automatic feature extraction from URLs

- Use of advanced models like Random Forest
- Development of a web-based interface

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11. CONCLUSION

This project demonstrates the application of machine learning in cybersecurity. The system effectively detects phishing websites and provides a simple interface for user interaction. It highlights the importance of AI in solving real-world problems.

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12. LEARNING OUTCOMES

- Understanding of classification algorithms
- Hands-on experience with ML pipeline
- Data preprocessing and evaluation
- Real-world problem solving

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13. REFERENCES

- Kaggle Dataset: Phishing Website Dataset
- Scikit-learn Documentation
- Python Documentation

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14. Images

Code:

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
import joblib
import os

def run_system():
    print("========================================")
    print("   PHISHING WEBSITE DETECTION SYSTEM")
    print("   Course: Fundamentals of AI and ML")
    print("   Name: Sarthak Jindal")
    print("   Reg No: 25BCE10189")
    print("========================================")
    print("\nWelcome! This AI model classifies URLs.")
    csv_file = 'Phising_Training_Dataset.csv'
    if not os.path.exists(csv_file):
        print(f"Error: {csv_file} missing.")
        return

    print("\nLoading dataset...")
    df = pd.read_csv(csv_file)
    print("Training the AI model... please wait.")
    X = df.drop(['key', 'Result'], axis=1)
    y = df['Result']

    x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42)

    clf = DecisionTreeClassifier(criterion='entropy', max_depth=15)
    clf.fit(x_train, y_train)
    joblib.dump(clf, 'phishing_model.pkl')

    preds = clf.predict(x_test)
    acc = accuracy_score(y_test, preds)

    print("----------------------------------------")
    print(f" SUCCESS: Model Training Complete!")
    print(f" Final Accuracy Score: {acc * 100:.2f}%")
    print("----------------------------------------")

    while True:
        print("\n========================================")
        print("       INTERACTIVE LIVE DETECTOR")
        print("========================================")
        print("Enter values: 1 (Legitimate), 0 (Suspicious), -1 (Phishing)")

        try:
            # Asking for the top 6 most important features
            s_state    = int(input("1. SSL Final State: "))
            anchor     = int(input("2. URL Anchor Score: "))
            traffic    = int(input("3. Web Traffic Score: "))
            sub_domain = int(input("4. Having Sub-Domain: "))
            tags       = int(input("5. Links in Tags: "))
            prefix     = int(input("6. Prefix-Suffix (Hyphen in Domain): "))

            # Creating a full 30-feature row for the model
            test_input = [0] * 30

            # Mapping inputs to their correct column positions in the dataset
            test_input[7]  = s_state
            test_input[13] = anchor
            test_input[25] = traffic
            test_input[6]  = sub_domain
            test_input[14] = tags
            test_input[5]  = prefix

            final_df = pd.DataFrame([test_input], columns=X.columns)
            res = clf.predict(final_df)[0]

            print("\n********************")
            if res == 1:
                print("RESULT: Website is SAFE.")
            elif res == 0:
                print("RESULT: Website is SUSPICIOUS.")
            else:
                print("ALERT: This is a PHISHING site!")
            print("********************")

        except ValueError:
            print("Invalid input. Use -1, 0, or 1.")

        choice = input("\nCheck another site? (y/n): ").lower()
        if choice != 'y':
            print("System shutting down. Stay safe online!")
            break

if __name__ == "__main__":
    run_system()
```

OUTPUT:

```
========================================
   PHISHING WEBSITE DETECTION SYSTEM
   Course: Fundamentals of AI and ML
   Name: Sarthak Jindal
   Reg No: 25BCE10189
========================================

Welcome! This AI model classifies URLs.

Loading dataset...
Training the AI model... please wait.
 SUCCESS: Model Training Complete!
 Final Accuracy Score: 95.48%

========================================
       INTERACTIVE LIVE DETECTOR
========================================
Enter values: 1 (Legitimate), 0 (Suspicious), -1 (Phishing)
1. SSL Final State: 1
2. URL Anchor Score: 0
3. Web Traffic Score: -1
4. Having Sub-Domain: 0
5. Links in Tags: 0
6. Prefix-Suffix (Hyphen in Domain): 1

********************
ALERT: This is a PHISHING site!
********************

Check another site? (y/n): y

========================================
       INTERACTIVE LIVE DETECTOR
========================================
Enter values: 1 (Legitimate), 0 (Suspicious), -1 (Phishing)
1. SSL Final State: 1
2. URL Anchor Score: 1
3. Web Traffic Score: 0
4. Having Sub-Domain: 0
5. Links in Tags: 0
6. Prefix-Suffix (Hyphen in Domain): -1

********************
RESULT: Website is SAFE.
********************
```

```
Check another site? (y/n): n
System shutting down. Stay safe online!
```

THANKYOU !